机器视觉与图像处理

Machine Vision & Image Processing

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Some clarification

- A distinction definition of IP
  - The input and output of the process is an image.
- Hyper Dictionary (www.hyperdictionary.com)
  - IP: Computer manipulation of images
  - CG: The pictorial representation and manipulation of data by a computer
  - PR: aims to classify data (patterns) based on either a priori knowledge or on statistical information extracted from the patterns.

- CV: A branch of artificial intelligence and image processing concerned with computer processing of images from the real world. It typically requires a combination of low level image processing to enhance the image quality and higher level PR and image understanding to recognize features present in the image. The ultimate goal is to use computers to emulate human vision, including learning and being able to make inferences and takes actions based on visual inputs.

- MV： Machine vision is the technology and methods used to provide imaging-based automatic inspection and analysis for such applications as automatic inspection, process control, and robot guidance, usually in industry. Machine vision refers to many technologies, software and hardware products, integrated systems, actions, methods and expertise. Machine vision as a systems engineering discipline can be considered distinct from computer vision, a form of computer science. It attempts to integrate existing technologies in new ways and apply them to solve real world problems. The term is the prevalent one for these functions in industrial automation environments but is also used for these functions in other environments such as security and vehicle guidance.

- The overall machine vision process includes planning the details of the requirements and project, and then creating a solution. During run-time, the process starts with imaging, followed by automated analysis of the image and extraction of the required information.

From Wikipedia, the free encyclopedia

Some clarification

- O. Faugeras, in preface of “Mathematical Problems in IP”
  - IP: the information is mostly the intensity values at the pixels, and the output is itself an image.
  - Image Analysis: the intensity values are enriched with some computed parameters, e.g., texture or optical flow, and by labels indicating such things as a region number or the presence of an edge; the output is usually some symbolic description of the content of the image.
  - CV, robot vision, machine vision: often use 3D information and perform some sort of abstract reasoning followed by decision-making and action.

Mathematics and IP

For more than two decades, the IP field was occupied mostly by computer scientists and electrical engineers. Its rather low level of mathematical sophistication reflected the kind of mathematical training that computer scientists and electrical engineers were exposed to. It is roughly limited to a subset of nineteenth-century mathematics.

Some heuristic methods can produce apparently startling results, but there is no precise characterization of why and when they work or don't work. The idea of proof of correctness of an algorithm under a well-defined set of hypotheses has long been almost unheard of in IP an IA.

O. Faugeras, in preface of "Mathematical Problems in IP"